Hetzner backup and local restoration

Claudio Verdi | APL | 24 October 2026

I restored a supplied server image in VirtualBox and verified that WordPress worked locally. On 24 October I checked the existing restoration using service status, HTTP tests and a browser. The local website returned HTTP 200 OK.

Scope and my contribution

The supervisor supplied the disk image and the completed wordpress-files.tar.gz and wordpress.sql backups. I did not create these backup files. My work covered downloading, decompression, conversion to VDI, local restoration and troubleshooting. Today's screenshots document checks of the existing files and the working virtual machine.

The assignment also asks for creation of a portable server backup. I did not perform that part myself and do not present it as completed. This report documents restoration from the supplied material.

Overview of approaches

A full disk image preserves partitions and the operating system together with installed software. RAW can be converted to VDI for VirtualBox. This is the approach used for this restoration.

A file backup together with an SQL export can restore WordPress in a prepared server environment, but does not itself contain a bootable server. Dedicated tools can also provide imaging and restoration; no such alternative was separately tested here.

Download and conversion

The WSL history records an SCP download from the root account at wordpress.multinomial.se into ~/hetzner-backup. The key hetzner_komvux was used. The file hetzner-backup-20260915-1244.img.gz was decompressed with gzip -dk, which keeps the compressed file.

qemu-img convert was used with raw input and vdi output. The destination was C:\Users\Claud\VirtualBox-Disks\wordpress_server_image.vdi. The history records the commands, but not their complete output. The existing VDI file and running machine were checked separately.

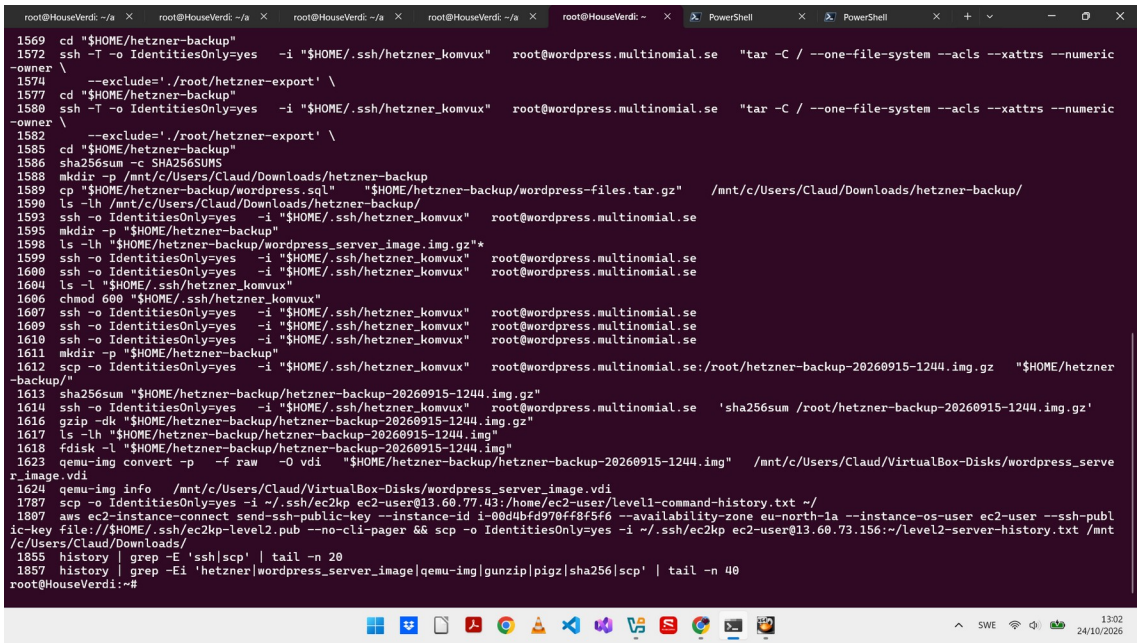


Figure 1. History of SCP, checksums, decompression and conversion

Difference from the reference guide

The guide uses wordpress_server_image.img.gz and the key hetzner_key. My history shows the dated image and the key hetzner_komvux. The report's commands use the actual filenames. Creation of the image using the guide's script is not presented as my work.

Disk image verification

The compressed image remains in WSL and is listed as 7.6G. gzip -t completed with an OK result. This verifies the gzip file's internal integrity, but does not replace comparison against an original SHA256 checksum.

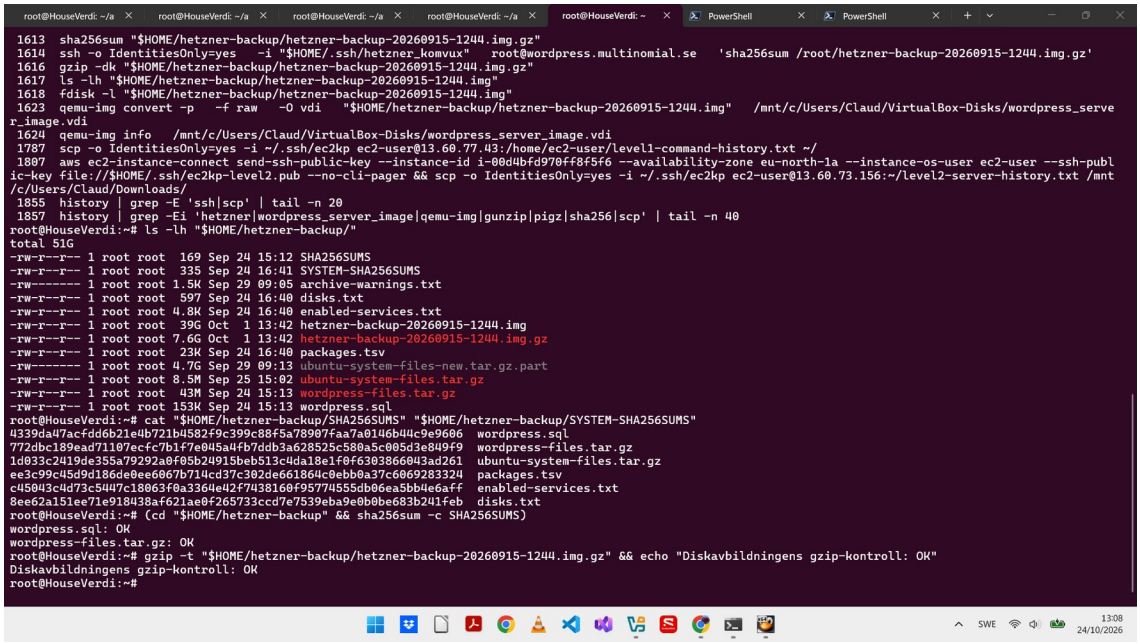


Figure 2. Successful gzip integrity check of the compressed disk image

fdisk -l reports 40,961,572,864 bytes, equivalent to 38.15 GiB, with GPT. The partitions include a 37.9G Linux filesystem, 1M BIOS boot and 256M EFI System. Their numbers are 1, 14 and 15. No partition table changes were made.

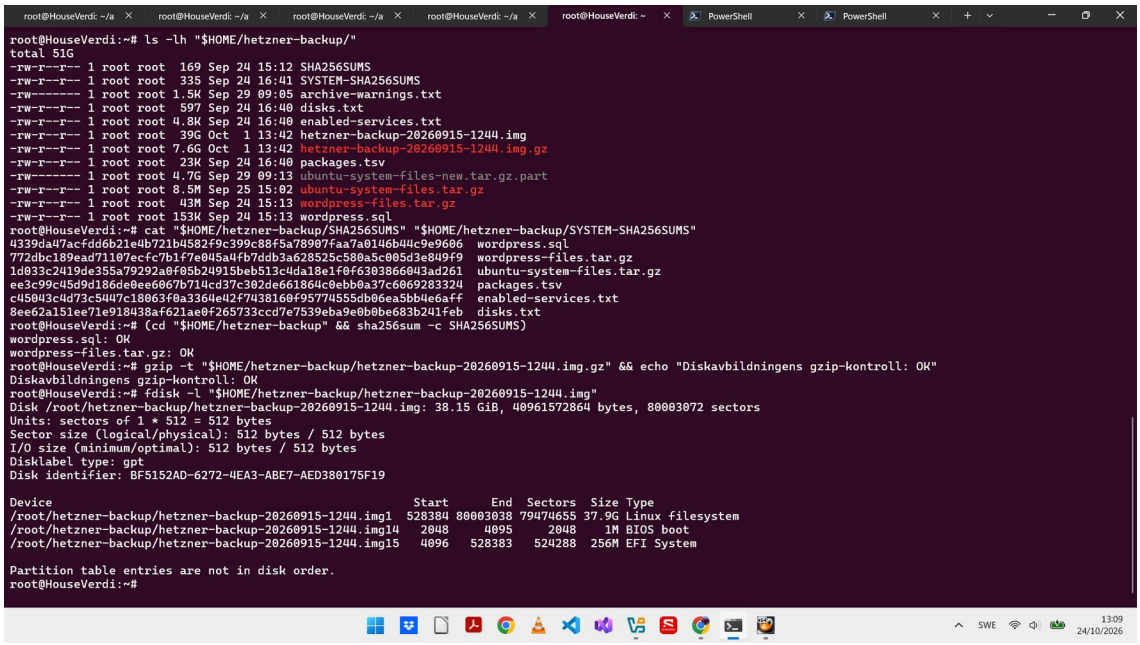


Figure 3. RAW image size and partitions

VirtualBox and the attached disk

Hetzner-Restore uses 4096 MB RAM, one CPU and BIOS firmware. Adapter 1 connects to the VirtualBox Host-Only Ethernet Adapter. Adapter 2 is disabled. This configuration provides local communication between Windows and the virtual machine.

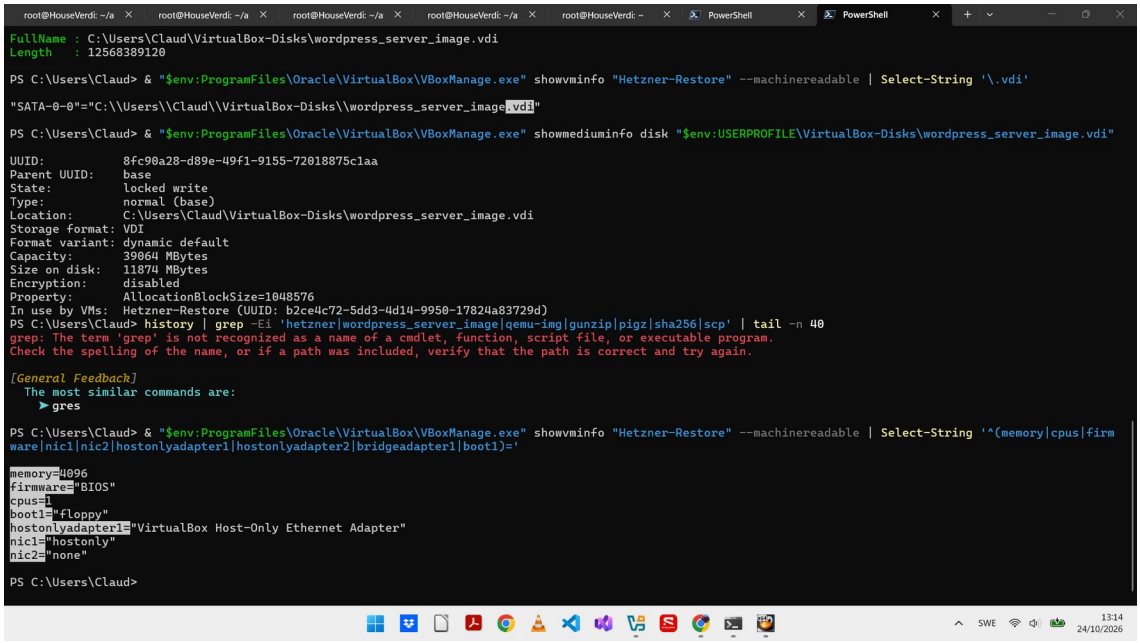


Figure 4. Confirmed memory, CPU, BIOS and network settings

The VDI is attached at SATA-0-0. showmediuminfo reports a dynamic VDI with a capacity of 39064 MBytes and that Hetzner-Restore uses it. The locked write state reflects its use by the running machine.

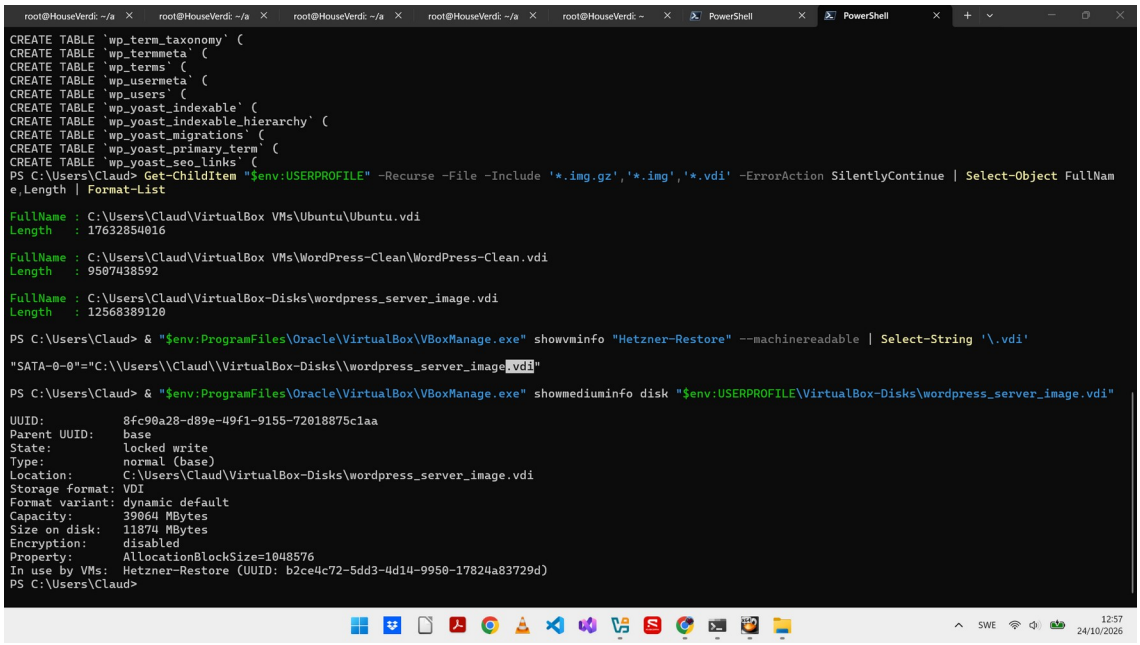


Figure 5. VDI format, capacity and association with Hetzner-Restore

Running server and services

The restored server boots Ubuntu 24.04.4 LTS with hostname komvux-ubuntu-4gb-hel1-2. Both apache2 and mysql returned active. This shows that the web server and database service were running at the time of the check.

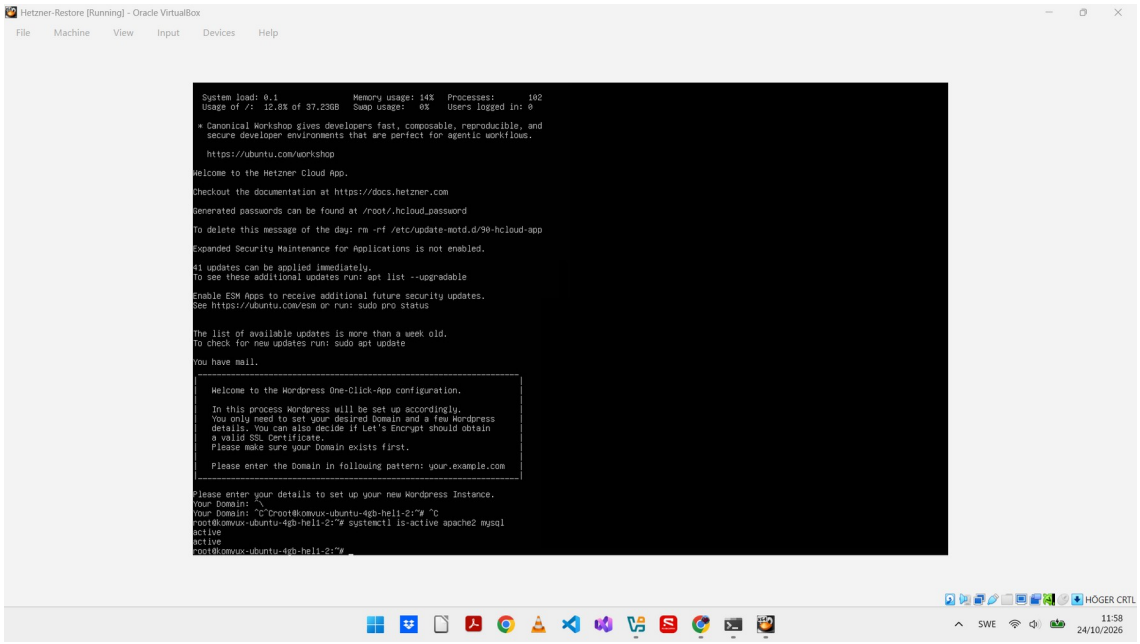


Figure 6. Apache and MySQL are active on the restored server

HTTP to localhost returned 301 and redirected to https://wordpress.multinomial.se/. A separate HTTPS test using --resolve to 127.0.0.1 returned 200 OK and a WordPress REST API link. The test ran inside the restored machine.

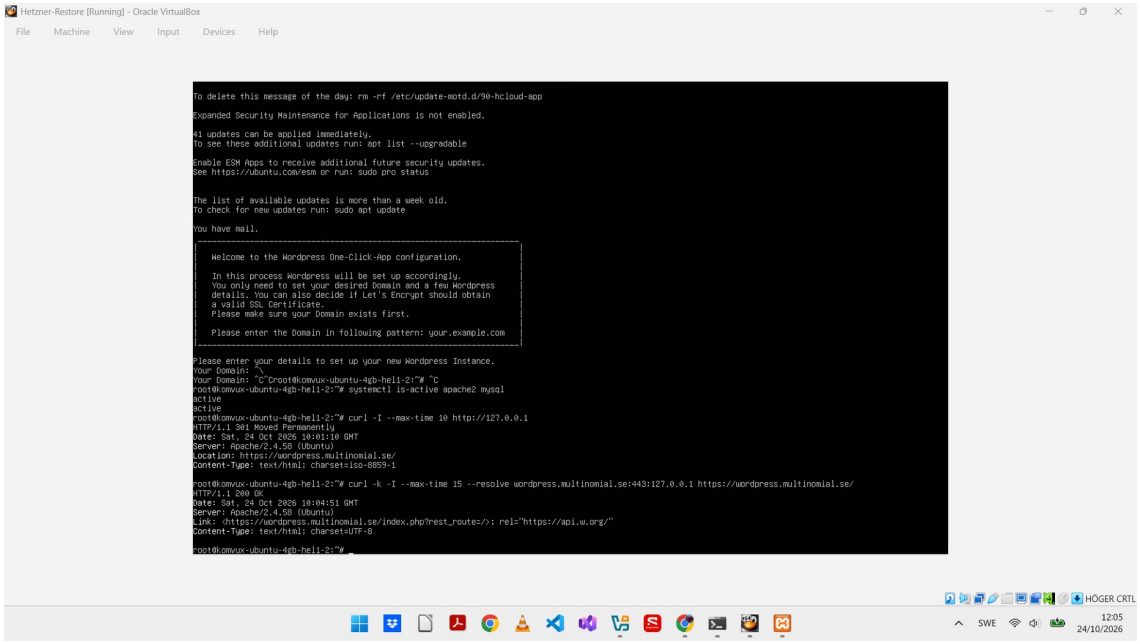


Figure 7. HTTPS 200 OK from the server's own local address

Local network and domain name

The server's enp0s3 interface has address 192.168.56.101/24. The Windows hosts file maps 192.168.56.101 to wordpress.multinomial.se. This allows the website's existing domain name to be used for the local restoration.

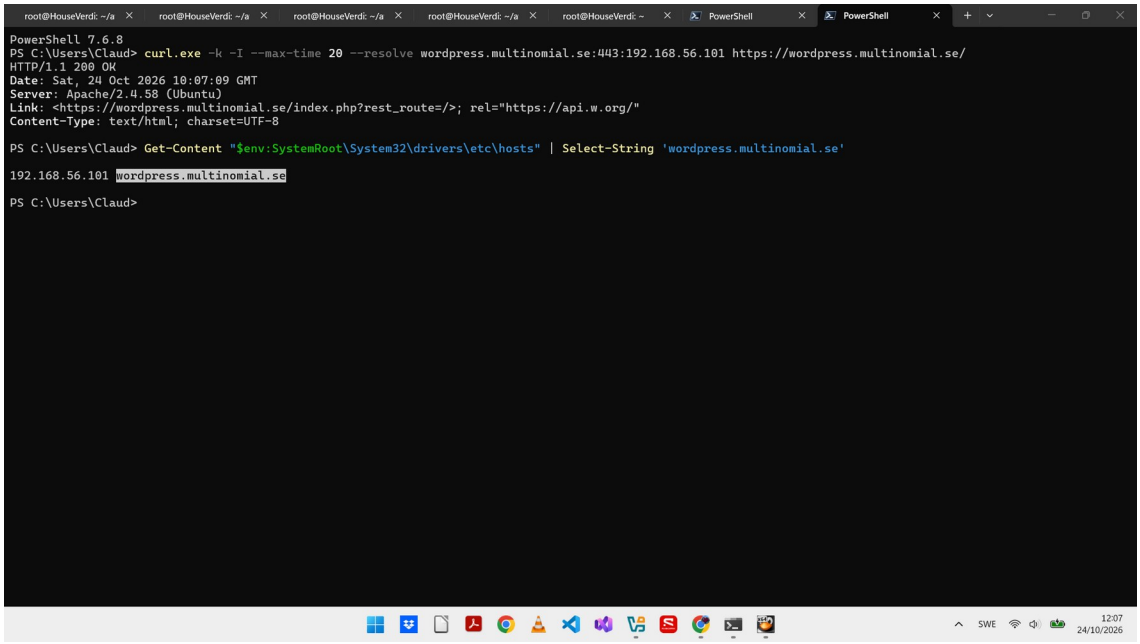


Figure 8. Windows hosts file mapping the domain to the local VM

From Windows, curl.exe with --resolve to 192.168.56.101 returned HTTP/1.1 200 OK. The response identified Apache/2.4.58 (Ubuntu) and text/html with UTF-8. The explicit IP address in the test shows that the response came from the local machine.

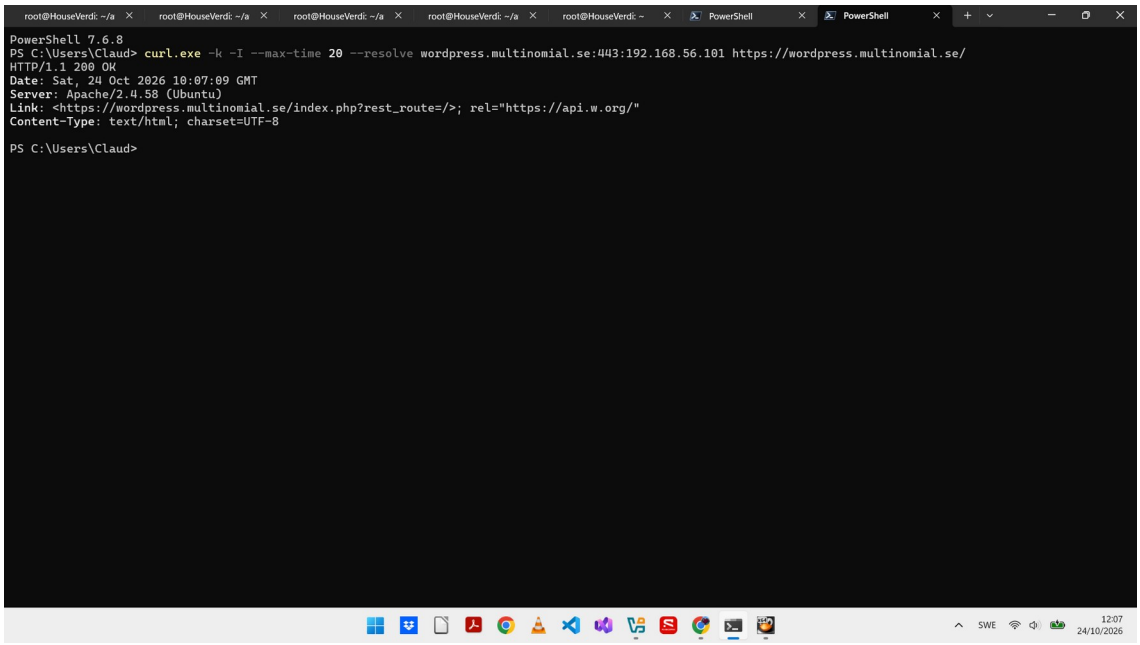


Figure 9. Windows reaches WordPress at the local VM address

Browser verification

The browser displays the website's front page with Website, Blog and the Hello world! post. This complements the HTTP test by showing that the page can also be rendered visually.

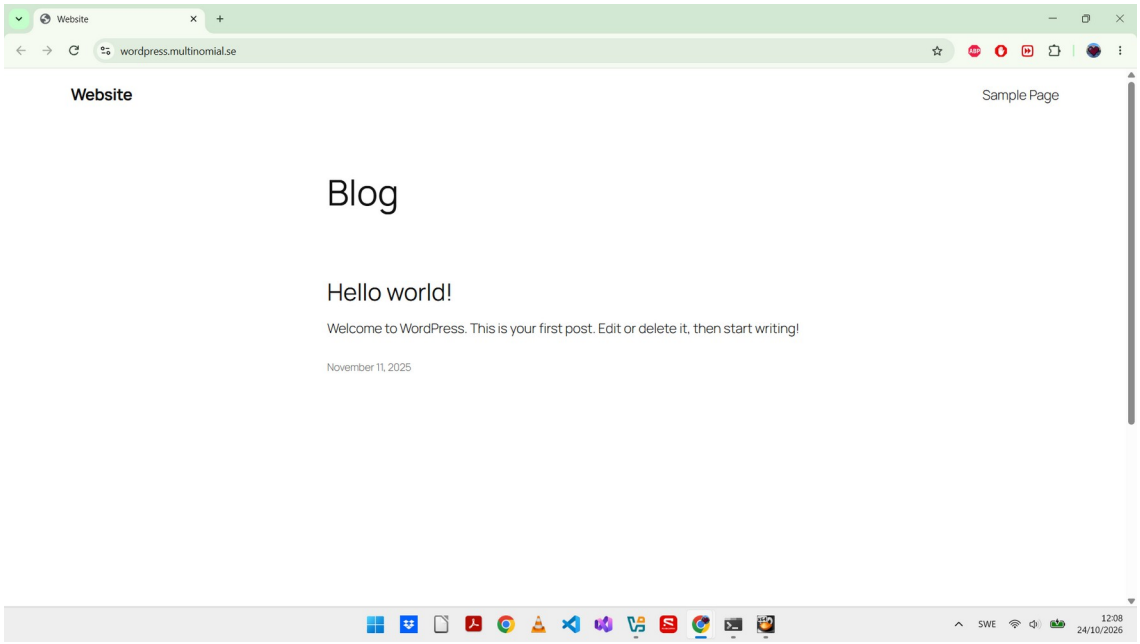


Figure 10. Front page of the restored WordPress website

The /wp-login.php address displays the WordPress login form. No login to the administration panel or content changes were documented during this check.

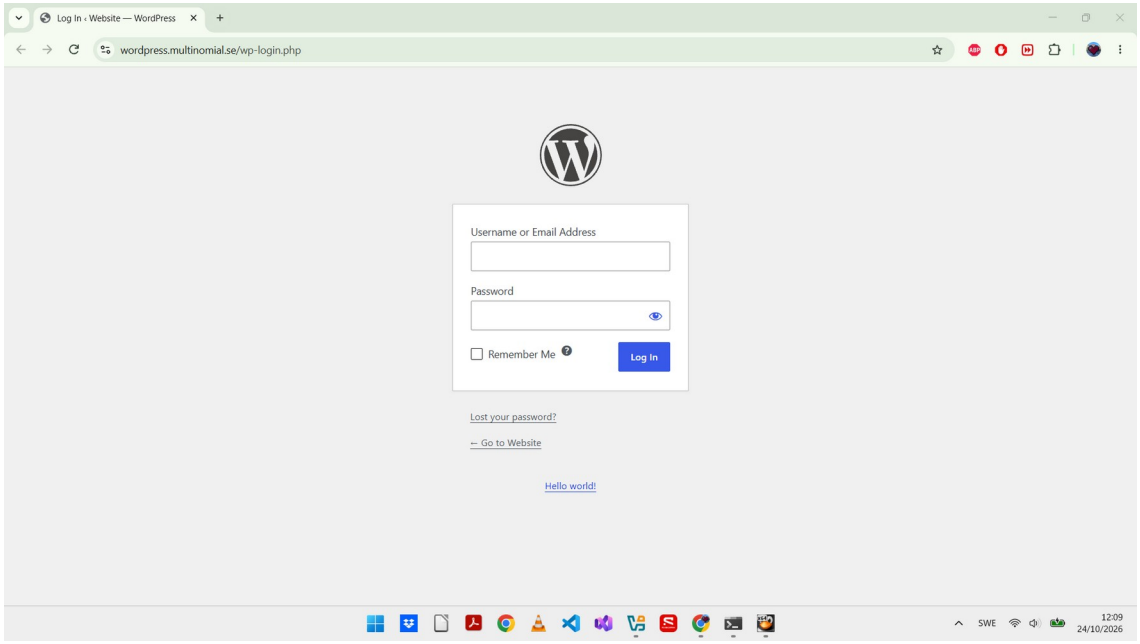


Figure 11. WordPress login form is displayed

Separate WordPress backups

The supplied files also exist in C:\Users\Claud\Downloads\hetzner-backup. The archive is 44,464,319 bytes and the SQL file is 155,699 bytes. These files complement the full disk image.

The tar listing shows the html directory including wp-config.php and wp-admin. This is a content listing; no new extraction or restoration from the archive was performed on 24 October.

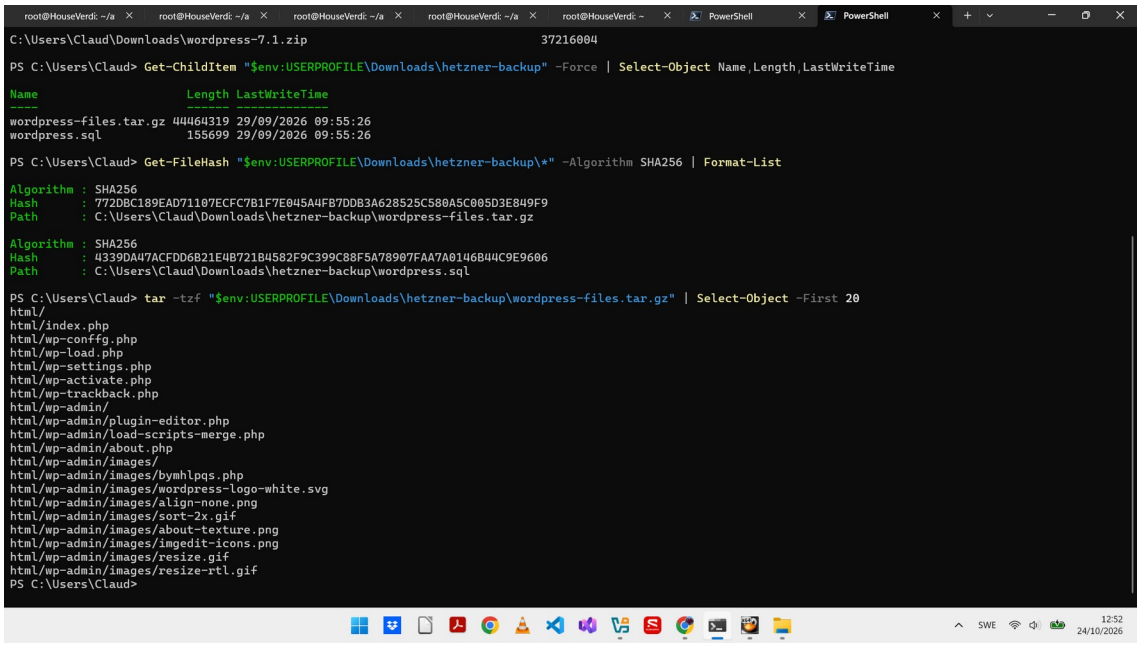


Figure 12. File listing from wordpress-files.tar.gz

The SQL file contains 17 CREATE TABLE statements, including WordPress and Yoast tables. Checking table names shows the file's structure, but is not a new import test.

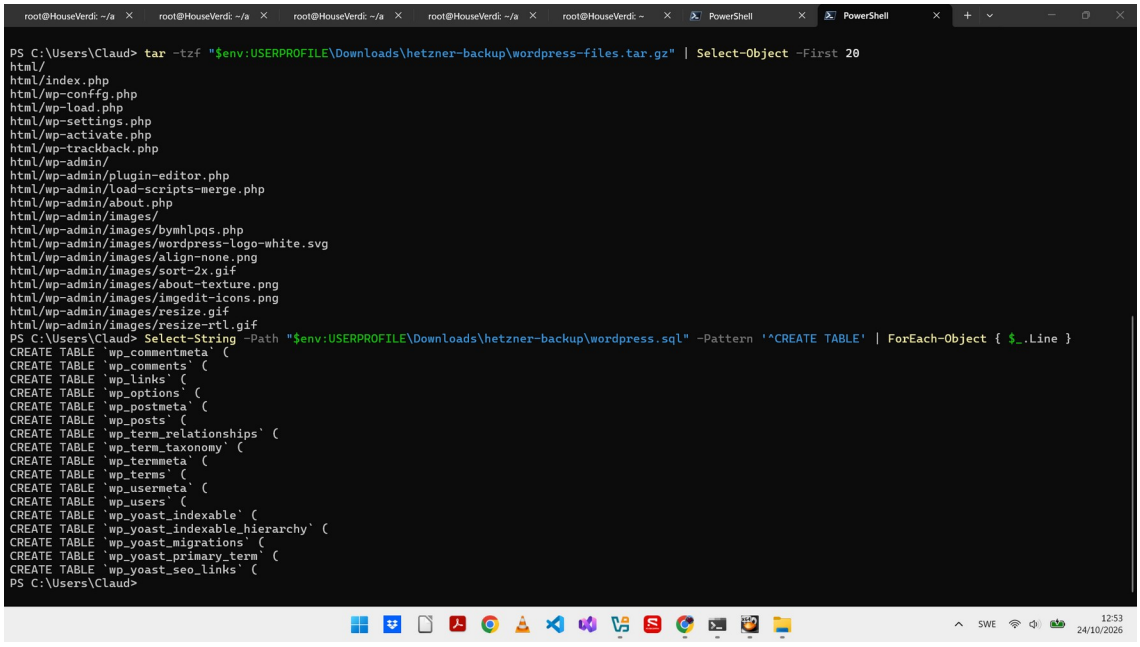


Figure 13. Table structure in wordpress.sql

Verified checksums

SHA256SUMS in WSL contains saved checksums for wordpress.sql and wordpress-files.tar.gz. sha256sum -c SHA256SUMS returned OK for both files. The values also match the checksums calculated for the Windows copies.

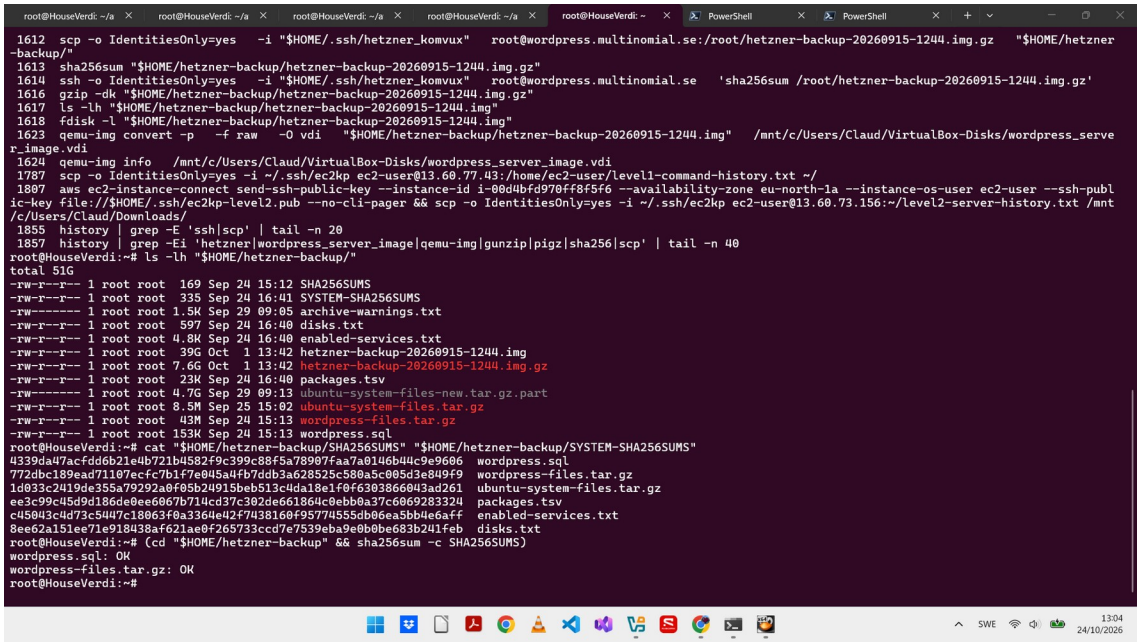


Figure 14. Both WordPress files match the saved SHA256 values

SHA256 for wordpress.sql

`4339da47acfdd6b21e4b721b4582f9c399c88f5a78907faa7a0146b44c9e9606`

SHA256 for wordpress-files.tar.gz

`772dbc189ead71107ecfc7b1f7e045a4fb7ddb3a628525c580a5c005d3e849f9`

SYSTEM-SHA256SUMS covers other system files and does not contain a checksum for the dated disk image. The history records local and remote SHA256 commands for that image, but their output is absent from the current evidence. Therefore, no newly verified comparison between its local and original hash is reported.

Troubleshooting and recovered access

During today's check I had forgotten the local virtual machine's login credentials. Access was recovered through GRUB using init=/bin/bash and the tty1 console. The root filesystem was remounted writable and the root password was changed with passwd root. The disk was then synced and the machine rebooted.

An initial attempt using the serial console produced a kernel panic. With the tty1 console I could continue. The account check showed root with Bash as its shell and an empty /home directory. I then logged in as root.

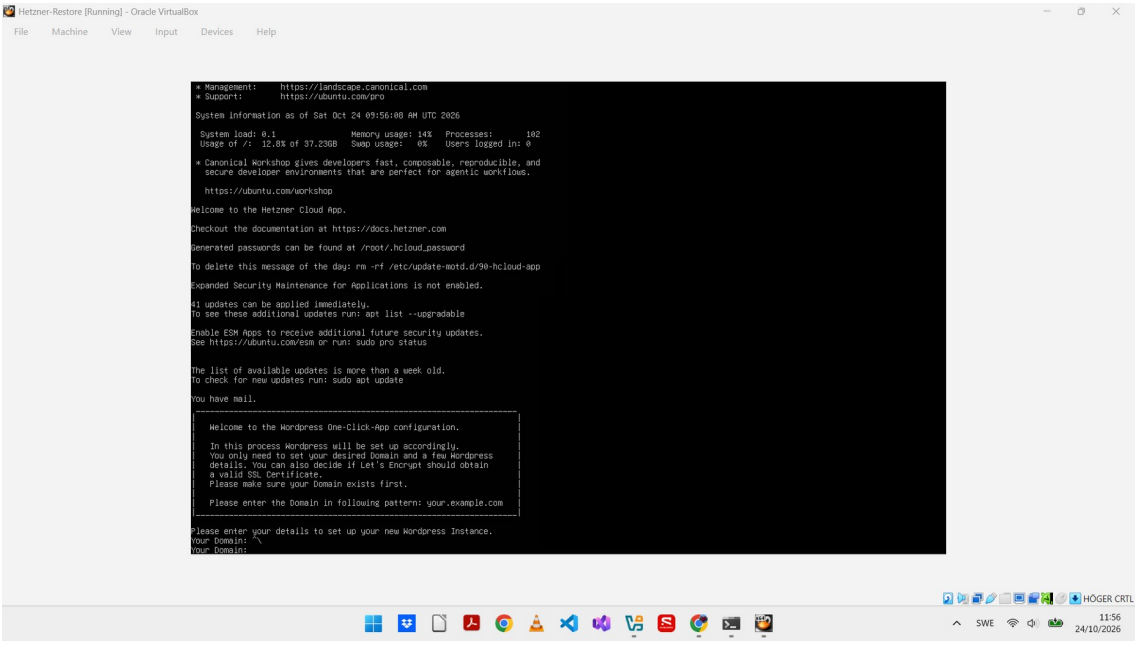


Figure 15. Successful root login after access recovery

After login, a WordPress configuration wizard started. I cancelled it with Ctrl+C to continue checking the existing installation. No new WordPress installation was performed at this step.

Workflow and improvements

PowerShell and Ubuntu/WSL use different syntax. When commands ran in the wrong terminal, switching tabs resolved the issue. For future work I should save terminal output alongside commands and checksums, so that each step can be traced beyond command history alone.

Commands and conclusion

The following commands reproduce the local decompression and conversion recorded in history and today's checks. Conversion has already been performed and must not target the VDI used by a running VM.

```
# Ubuntu / WSL
gzip -dk "$HOME/hetzner-backup/hetzner-backup-20260915-1244.img.gz"
qemu-img convert -p -f raw -O vdi \
  "$HOME/hetzner-backup/hetzner-backup-20260915-1244.img" \
  /mnt/c/Users/Claud/VirtualBox-Disks/wordpress_server_image.vdi
(cd "$HOME/hetzner-backup" && sha256sum -c SHA256SUMS)
gzip -t "$HOME/hetzner-backup/hetzner-backup-20260915-1244.img.gz"
fdisk -l "$HOME/hetzner-backup/hetzner-backup-20260915-1244.img"

# Ubuntu VM
systemctl is-active apache2 mysql
ip -br address
curl -k -I --max-time 15 \
  --resolve wordpress.multinomial.se:443:127.0.0.1 \
  https://wordpress.multinomial.se/

# Windows PowerShell
curl.exe -k -I --max-time 20 `
  --resolve wordpress.multinomial.se:443:192.168.56.101 `
  https://wordpress.multinomial.se/
```

The -k flag is used for this local connectivity check and disables TLS certificate verification. HTTP 200 shows that the website responds, but this test does not establish a valid certificate chain.

Result and remaining work

The supplied server image works as a local VirtualBox server with Apache, MySQL and WordPress. On 24 October the existing restoration was tested; no new full restoration was performed. Creation of my own portable backup and the video recording remain to be documented separately if these are to form part of the supplementary submission.

References: Task 01 - Hetzner Backup and Hetzner Backup Guide, supplied PDF files. Evidence: my embedded screenshots from 24 October 2026 and the visible terminal history. Private keys, passwords and SQL content containing user data are not reproduced.